

WTW 158 Worksheet 2 on Functions and Trigonometry

A. TRIGONOMETRY WORKSHEET on CLICKUP

Do before the prac

B. TEXTBOOK PROBLEMS - do before the prac

p 19 no 25, 34, 35, 43, 76

p A32: 20, 34, 68, 69, 74

C. MORE PROBLEMS

1.1 Sketch an **odd** function f such that $f(x) = \begin{cases} 2 & \text{if } 0 < x \leq 1 \\ 2/x & \text{if } 1 < x \leq 3 \end{cases}$

1.2 Find the domain and range of f .

2. Find (i) $\sin 50\pi$ (ii) $\cot \frac{15\pi}{4}$ (iii) $\sec(-\frac{11\pi}{6})$ (iv) $\operatorname{cosec} \frac{11\pi}{2}$

3. Solve for x :

(i) $\sin^2 x = 1, x \in [0, 2\pi]$

(ii) $2\sin^2 x = 3\cos x, x \in [-\pi, \pi]$

4. Solve by means of a graph: $\sin x > \cos x, x \in [0, 2\pi]$

D. TEST QUESTIONS

Question 1

Write f out as a piece-wise defined function and determine the domain and range of the function $f(x) = |1 - 4x^2|$.

Question 2

$\sin(-\frac{13\pi}{2}) =$

[a] 1	[b] -1	[c] 0	[d] None of these
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Question 3

If $\sin x = -\frac{1}{3}, \pi < x < \frac{3\pi}{2}$, then $\cot x =$

[a] $\sqrt{8}$	[b] $-\sqrt{8}$	[c] $\frac{1}{\sqrt{8}}$	[d] $-\frac{1}{\sqrt{8}}$	[e] $\frac{3}{\sqrt{8}}$	[f] $-\frac{3}{\sqrt{8}}$	[g] None of these
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Question 4

Solve for x if $2\cos x \sin x = -\cos x$.

Question 5

Determine $x \in [0, 2\pi]$ such that $|\cos x| \leq \frac{1}{2}$ by using a sketch.

E YOUTUBE VIDEOS

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A way to remember the entire unit circle for trigonometry

Solving trigonometric equations

SELECTED ANSWERS

A. Solutions on ClickUP

B. p19 34. $t \in [-2, 3]$ 35. $x \in \mathbb{R} \setminus [0, 5]$ 76. f is odd. p32 68. $x \in \{\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}\}$,
69. $x \in \{\frac{\pi}{6}, \frac{5\pi}{6}, \frac{\pi}{2}, \frac{3\pi}{2}\}$ 74. $x \in [0, \frac{2\pi}{3}) \cup (\frac{4\pi}{3}, 2\pi]$

C. 1.2 Domain: $[-3, 3] \setminus \{0\}$ Range: $[-2, -\frac{2}{3}] \cup [\frac{2}{3}, 2]$ 2 (i) 0 (ii) -1, (iii) $\frac{2}{\sqrt{3}}$ (iv) -1

3.(i) $x \in \{\frac{\pi}{2}, \frac{3\pi}{2}\}$ (ii) $x = \{-\frac{\pi}{3}, \frac{\pi}{3}\}$ 4. $x \in (\frac{\pi}{4}, \frac{5\pi}{4})$

D. Q1: Dom $\mathbb{R}$, Range $[0, \infty)$ Q2: [b], Q3: [a] Q4: $x = \frac{\pi}{6} + k\pi, x = \frac{5\pi}{6} + 2k\pi, k \in \mathbb{Z}$,

Q5. $x \in [\frac{\pi}{3}, \frac{2\pi}{3}] \cup [\frac{4\pi}{3}, \frac{5\pi}{3}]$